

Pump

Syringe pump x 3 (Chemyx: Fusion 100)

Syringe (Norm-ject plastic)	Min flow rate (mL/min)	Max flow rate (mL/min)	Max Pressure
1 mL	0.0001	3.2452	<3 bar
2 (3) mL	0.0001	13.8227	
5 (6) mL	0.0001	22.8812	
10 (12) mL	0.0001	37.0935	
20 (24) mL	0.0001	58.7368	

Pump

Peristaltic pump (Vapourtec: SF-10, E-series)

Flow rate: 0.02 mL/min – 10 mL/min

Max pressure: <10 bar (recommend <8 bar)

HPLC pump (Vapourtec: R-series, R2C+ pump)

Flow rate: 0.01 mL/min – 10 mL/min

Max pressure: <42 bar

Tubing Reactor

0.04" (0.10 mm) ID

- 2 mL x 1 (PFA)
- 5 mL x 2 (PFA)
- 10 mL x 2 (PFA, FEP)
- 20 mL x 1 (FEP)

0.03" (0.10 mm) ID

- 5 mL x 1 (PFA)

Max pressure : <34 bar

Max temperature : FEP: 50 °C / PFA: 80 °C

Photo reactor

UV-150 photo reactor (Vapourtec)

Wavelength (nm)	Input power	Radiant power	Wavelength (measured)	Optical power (mW/cm ²)
365	60 W	3 W	-	-
380	60 W	4.2 W	-	-
395	60 W	6 W	396-	-
420	60 W	18 W	424	243
450	60 W	24 W	450	355
470	60 W	24 W	-	-
525	60 W	3 W	533	-

temperature : 40 – 80 °C

Manual photo reactor

Wavelength (nm)	Input power	Radiant power	Wavelength (measured)	Optical power (mW/cm ²)
Blue	24 V	-	448	62

temperature : 40 or 50 °C

MFC

Gas	Inlet pressure	Outlet pressure	Flow range
O ₂	10 bar	9 bar	0.01 – 10 sccm

Hydrogenation reactor



ThalesNano H-cube mini plus

Temperature range	25 – 100 °C
Pressure range	1 – 100 bar
Flow rate	0.1 – 3.0 mL/min
Hydrogen Generation	25 – 30 mL/min
Water Reservoir Capacity	100 mL
Catalyst Amount	0.1 – 0.3 g
Concentration	0.01 – 1.0 M